

PROJECT REPORT

SyncCampus

Dynamic Academic Timetable and Lecture Communication System

Change the timetable once, and everyone affected sees the correct update.

Proposed design for a first prototype

Web-based system for colleges: Students, Teachers, Admin / HOD

1. Introduction

College timetables are prepared once, at the start of a semester. Real schedules do not stay that fixed: lectures get cancelled, moved to another time, or shifted to another room. These changes are normally announced separately from the timetable, so students keep following a timetable that is no longer correct.

SyncCampus is a web-based system that makes the timetable itself dynamic. A teacher changes a lecture from a personal timetable; the system checks for clashes, updates the official timetable, and every affected student sees the new version live.

This report explains the problem, why existing methods do not fully solve it, the proposed solution, its design, and the scope for future work. SyncCampus is currently a proposed design; no implementation results are claimed here.

	Summary
Problem	The official timetable and the real daily schedule drift apart.
Root cause	Changes are communicated separately; the timetable never changes.
Solution	A live timetable that teachers and admins update, with automatic clash checking.
Users	Students (view only), Teachers (own lectures), Admin / HOD (whole timetable).
Key idea	Weekly timetable + one-day exceptions. The timetable is the source of truth.

2. Problem Statement

Problem statement

College timetables are fixed, but real academic schedules change often. Because a change is communicated separately from the timetable, the two stop matching. Students who trust the timetable go to the wrong room or attend a lecture that was cancelled, and teachers must pass the same news to every class they teach.

2.1 What is actually going wrong

The real problem is not that nobody tells students. Messages usually are sent. The real problem is **disconnection**: the information lives in two places. The timetable says one thing, a message says another, and the student has to combine both in their head and remember which is newer.

Example: the timetable says DBMS, 9:00 AM, Room 201. At 8:15 AM the teacher posts "today's lecture is cancelled" in a group. A student who has already left home, or does not open the group, still arrives at Room 201.

TODAY: timetable and changes travel separately

The official timetable never changes. The truth lives in messages.

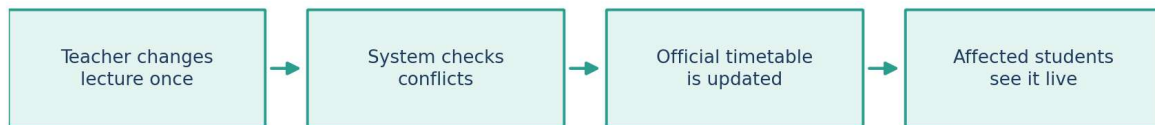
WITH SYNCCAMPUS: the timetable itself changes

Figure 1. Today's process compared with the SyncCampus process

2.2 Who is affected

Group	Problems faced
Students	Follow an outdated timetable Miss a cancellation or room change Reach the wrong classroom or arrive late Cannot plan free time (for example, a cancelled 9:00 lecture before a 10:00 lecture)
Teachers	Teach several classes, divisions or departments Must find and inform each affected group separately Repeat the same message for every change
Admin / HOD	Manage teachers, classes, subjects, rooms and the timetable in separate places No single record of who changed what, and when

2.3 What a web search shows

Searching for how colleges handle this gives a consistent picture:

- A 2026 college notice names WhatsApp groups as the main platform for daily timetables, lecture links and notices [1].
- A university project thesis reports that notice-board timetables are often late or inaccurate because they are built by hand, and that students miss classes and end up with short attendance as a result [2].
- SMS-alert providers describe the same scene: students learn a lecture is cancelled only after they reach the lecture hall [3].
- A UK students' union ran a text-message campaign to collect reports of lectures cancelled without notice [8].

3. Existing Solutions and Their Problems

Colleges currently rely on a mix of the methods below. Several address part of the problem, but most keep the timetable and the news about changes in different places.

Existing solution	How it works	Where it falls short
Notice board, PDF or Excel timetable	A fixed timetable is shared once at the start of the semester.	Cannot show a one-day change. Manual preparation causes errors [2]. Updates must be found elsewhere.
WhatsApp / messaging groups	A teacher or class representative posts in each class group [1].	Separate from the timetable. Messages get buried. The teacher must find every affected group. No change record and no clash check.
Email or SMS alerts	A bulk message is sent when a lecture is cancelled [3].	Still only a message; the timetable is unchanged. Depends on correct contact lists. Alerts may be late or unread.
College ERP / student portal	A central system holds the timetable and other records [6].	In many deployments: broad multi-module systems, timetable edits go through the admin, and adoption needs a college-wide rollout.
Timetable software, e.g. OpenEduCat	Generates timetables and notifies affected users by portal, email or SMS [4].	Closest to our idea. Built as a full suite centred on timetable generation, which is more than a department needs for daily changes.
University timetable and reminder apps	Show a personal timetable and alert on room changes [5][7].	Rely on the institution's own back-end or on data entered by hand. Not built around a teacher editing their own lecture with a clash check.

3.1 Comparison

The table below is a general comparison of typical setups; individual products vary.

Approach	Timetable updates itself	Teacher edits own lecture	Clash check	Only affected class informed	Light to adopt
Notice / PDF / Excel	No	No	No	No	Yes
WhatsApp groups	No	No	No	Partly	Yes
Email / SMS	No	No	No	Partly	Yes
Full ERP / portal	Yes	Partly	Partly	Yes	No
SyncCampus (proposed)	Yes	Yes	Yes	Yes	Yes

The gap

Systems that update the timetable and notify students do exist, but they are large platforms. Many colleges still run on a static timetable plus messaging groups. There is room for a small, focused system where the teacher edits their own lecture, clashes are checked automatically, and a one-day change never disturbs the permanent timetable.

4. Proposed Solution: SyncCampus

SyncCampus connects the timetable, teachers, students and administrators in one system. Its guiding principle is:

The timetable is the source of truth. A change is not just announced; the timetable itself is updated.

4.1 Weekly timetable plus one-day exceptions

The timetable repeats every week. When a teacher cancels or moves a lecture, the change applies to that one date only. The system keeps the permanent weekly slot untouched and stores the change as an exception for that date. What a student sees for any date is the weekly slot combined with any exception for that date.

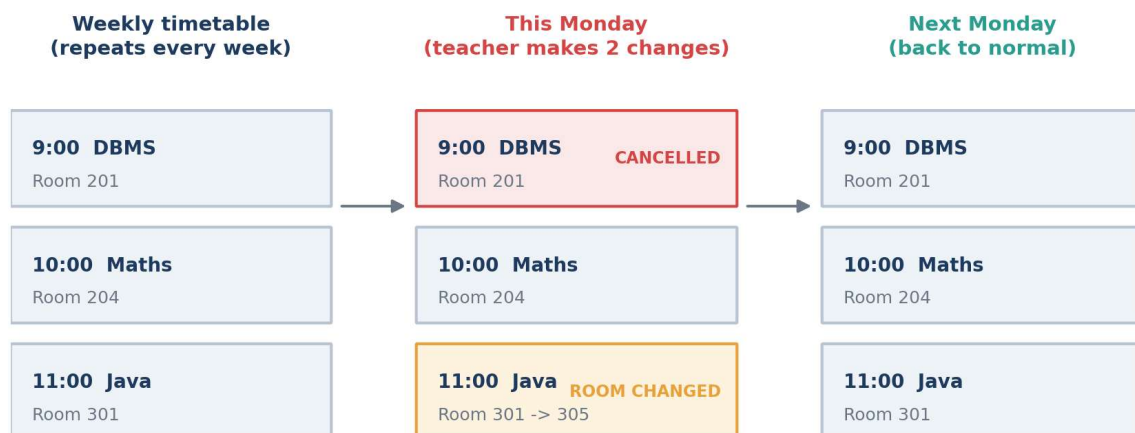


Figure 2. A one-day change; the following week returns to normal automatically

4.2 Users and permissions

Role	Can do	Cannot do
Student	Log in; view daily and weekly timetable; see live changes (cancelled, rescheduled, room changed).	Change anything in the timetable.
Teacher	View a personal timetable of own lectures; cancel, reschedule or change the room of own lectures.	Change another teacher's lectures.
Admin / HOD	Create and change the timetable according to teacher and room availability.	Save a change that causes a clash.

4.3 How it works

1. The admin creates the weekly timetable for each division or batch.
2. Students and teachers log in and see their own timetable only.
3. A teacher selects a lecture and chooses Cancel, Reschedule or Change Room.
4. The system checks permission and clashes (Section 4.4).
5. If valid, the change is saved as a one-day exception and recorded in the change history.
6. The system identifies the affected division or batch and updates their timetable live.

Example

Professor A teaches FY A (DBMS, 9:00), FY B (DBMS, 11:00) and SY A (Java, 2:00). She cancels the 9:00 lecture. Only FY A's timetable shows it as cancelled. FY B and SY A are unaffected, and next week FY A's 9:00 DBMS lecture appears as usual.

4.4 Conflict management

Before any change is saved, the system checks three things:

- **Teacher:** the teacher must not already be teaching at the new time.
- **Room:** the room must not already be booked at the new time.
- **Division / batch:** the class must not already have a lecture at the new time.

If any check fails, nothing changes. The user sees a clear message and must choose another room or time.

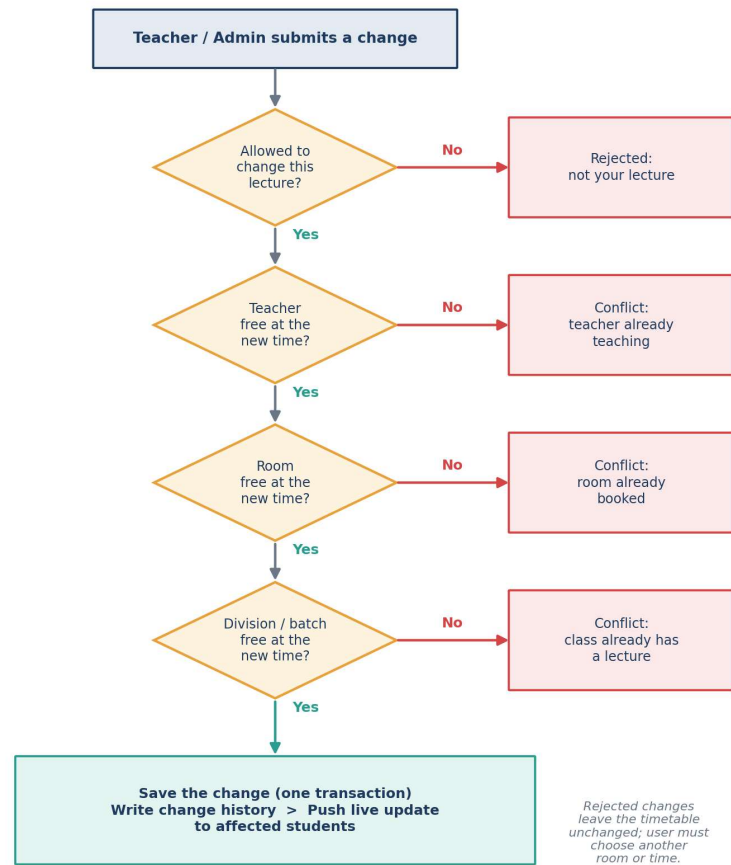


Figure 3. Checks performed before a change is saved

4.5 Two teachers, same room, same time

Checking in the user's screen is not enough. Two teachers may both see Room 305 as free and submit at nearly the same moment. The final decision is therefore made by the database: it handles the requests one at a time inside a transaction, saves the first valid one, and rejects the second with a clear message.

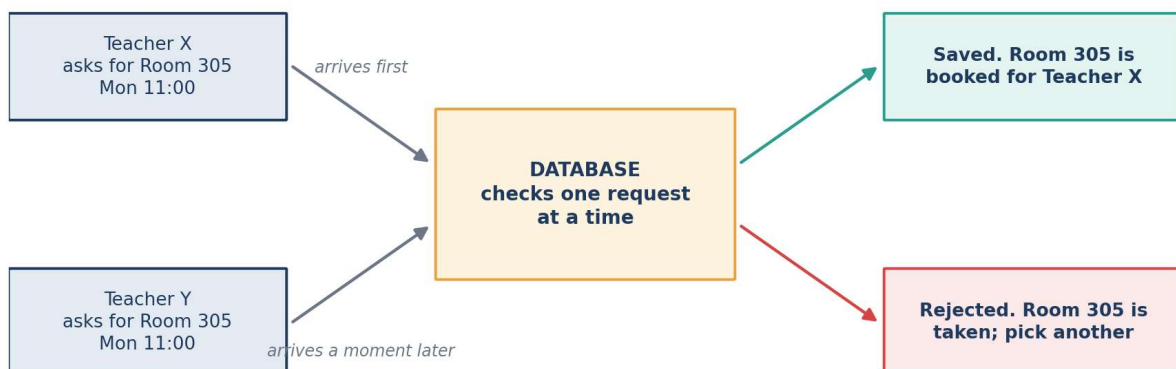


Figure 4. The database makes the final decision when two requests collide

5. Prototype Scope

The first prototype is deliberately small. It contains only the features below.

Included in the prototype	Not included (see Future Scope)
Login with role-based access Student: read-only daily and weekly timetable Teacher: personal timetable; cancel, reschedule, change room Admin / HOD: create and change the timetable Weekly timetable with one-day exceptions Live updates without refreshing Conflict management (teacher, room, class / batch) Change history	Extra lectures, holidays and examinations Push notifications Substitute-teacher workflow Attendance, analytics, calendar or WhatsApp integration Automatic timetable generation

6. System Design

6.1 Academic structure

Colleges organise students in layers. Some classes have only divisions; others also have batches (for example, for practicals). SyncCampus follows this hierarchy so that a change reaches exactly the right students.



A lecture targets a whole Division (theory) or a single Batch (practical).

Figure 5. Academic hierarchy; the batch level is optional

A theory lecture is linked to a whole division and a practical to a single batch. A change to a batch practical therefore affects only that batch.

6.2 Architecture

SyncCampus uses a three-tier architecture: a frontend for users, a backend that holds the rules, and a database that stores the data and makes the final conflict decision.

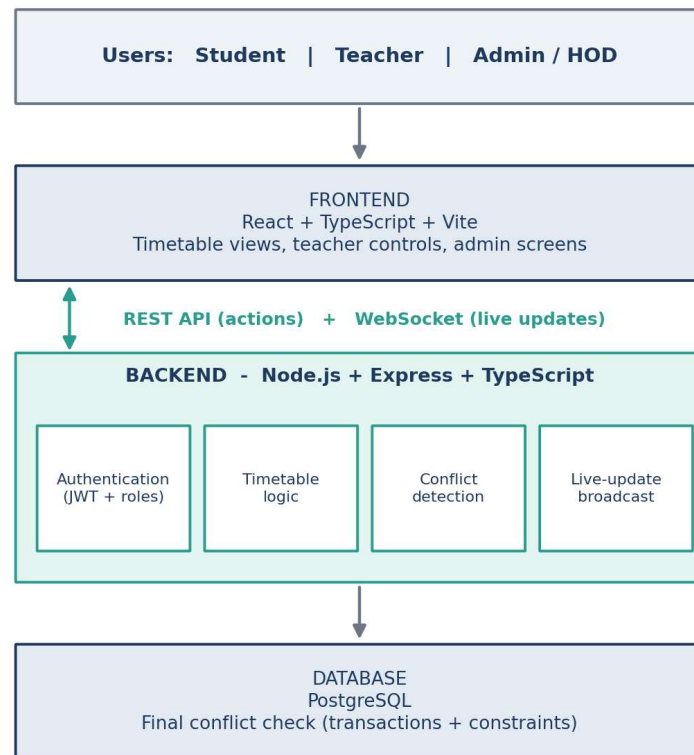


Figure 6. Three-tier architecture

6.3 Technology

Layer	Choice	Reason
Frontend	React + TypeScript + Vite	Suited to dynamic, role-based timetable screens.
Backend	Node.js + Express + TypeScript	REST API, business rules and conflict checks in one language.
Database	PostgreSQL	Strongly linked data; transactions and constraints keep the timetable consistent.
Live updates	WebSocket (Socket.IO)	Pushes a change to open timetables without refresh.
Security	JWT login, hashed passwords, role checks	Students read only; teachers edit only their own lectures.

6.4 Data model

Table	Main fields	Purpose
users	id, name, email, password_hash, role	Login and roles.
departments, courses, divisions, batches	ids, names, year; batch is optional	Academic hierarchy.
students, teachers	id, user_id, division / batch or department	People linked to the hierarchy.

Table	Main fields	Purpose
subjects_rooms	id, name; room number, type, capacity	What is taught and where.
weekly_timetable	subject, teacher, division, batch (optional), room, weekday, start, end	The permanent repeating slots.
lecture_exceptions	weekly slot, date, status (cancelled / changed), new time, new room, changed_by	A change for one date only.
change_history	exception, changed_by, old value, new value, time	Who changed what, and when.

Because one exception can carry a new time and a new room together, a lecture can be rescheduled and moved at once without extra status types.

7. Development and Testing Plan

The system will be built in small phases, with the riskiest part (conflicts) tested before the live layer is added.

Phase	Work	Result
1	Database and login with roles	Users can sign in to the right dashboard.
2	Weekly timetable and views	Admin creates it; students and teachers view it.
3	Lecture changes	Cancel, reschedule and change room as one-day exceptions.
4	Conflict detection	Teacher, room and class / batch clashes are blocked.
5	Live updates	Open timetables update without refresh.
6	Testing	All scenarios below pass.

Planned test scenarios:

- A teacher cancels a lecture: only that date changes; the next week is normal.
- A teacher tries to change someone else's lecture: rejected.
- A student tries to edit: not possible.
- Reschedule into a time the teacher is already teaching: rejected with a message.
- Two teachers request the same room and time together: exactly one succeeds.
- A change to one batch practical: only that batch's timetable changes.

8. Expected Benefits and Limitations

8.1 Benefits

For	Benefit
Students	Always see the latest timetable; fewer wrong-room trips; better planning of free time.
Teachers	Change a lecture in one place; no hunting for class groups; no repeated messages.
Admin / HOD	One central timetable; clash-free changes; a clear record of every change.
College	One reliable source for the current schedule, less dependence on informal messages.

8.2 Limitations

- The system is only as correct as the data entered. A wrong original timetable will be shown wrongly.
- The prototype updates the timetable live, but without push notifications a student sees a change when they open the app.
- The prototype covers one-day changes to regular lectures only. Holidays, exams and extra lectures are not handled yet.

9. Future Scope

These ideas are deliberately kept out of the first prototype.

Future feature	What it would add
Extra lectures, holidays, examinations	Show these on the timetable alongside regular lectures.
Push notifications	Alert only affected students on their device when a change is made.
Substitute teacher	Admin assigns a replacement teacher for a given date.
Free-room suggestions	Suggest available rooms when a lecture must move.
Timetable conflict reports	Help the admin find clashes while building the timetable.
Mobile app	A dedicated app for faster everyday use.
Analytics	Cancelled and rescheduled lectures, room usage, department activity.

10. Conclusion

Timetables are static while real schedules are not, and the gap is filled today by messages that students must remember. SyncCampus closes that gap by making the timetable itself the place where changes happen. The teacher changes a lecture once, the system checks for clashes, and the right students see the update live. By staying small and focused, with a weekly timetable, one-day

exceptions, role-based access and strict conflict checking, the first prototype targets the core problem without extra complexity.

Change once > Validate > Update the timetable > Students see it live

References

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